The National Higher School of Artificial Intelligence
Introduction to Artificial Intelligence
Final Exam 25/05/2025
Duration: 2 hours

Last Name	
First Name	
Group #	

Exercise 1		**/ 21**
Exercise 1 (scaled down)		**/ 6**
Exercise 2		**/ 7**
Exercise 3		**/ 8**
TOTAL **Potential 1pt bonus**		**/21**

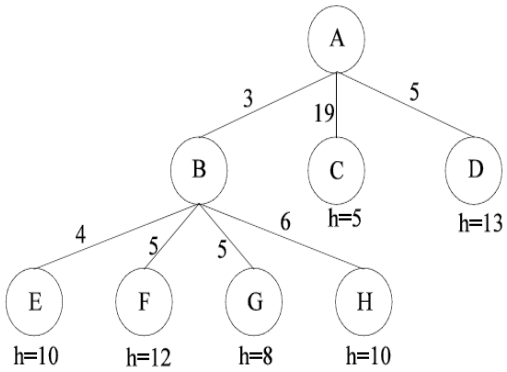
1. **ANSWER ON THESE EXERCISE STATEMENT PAGES. IF YOU DO NOT HAVE ENOUGH, SPACE WRITE ON THE BACK OF THE SAME PAGE.**
2. **Use a PEN NOT a PENCIL everywhere. (A penalty will be applied if you use a pencil.)**
3. **DO NOT** USE ANY RED OR PINK (OR RELATED) COLOUR IN YOUR ANSWERS**!**
4. **FILL IN ALL THE IDENTIFICATION BOXES OTHERWISE PAGES MAY GET MIXED UP.**
5. **WRITE NEATLY SO AS NOT TO BE PENALISED!**

Exercise 1: 6 points

In the following, circle THE correct answer. **No penalty for wrong guesses.**

REPORT ALL YOUR ANSWERS TO THIS EXERCISE QUESTIONS ON THE ANSWER SHEET GIVEN ON PAGE 5.

1. Consider the following *search tree* produced after expanding nodes A and B, where each arc is labelled with the cost of the corresponding operator, and the leaves are labelled with the value of a heuristic function, *h*. For uninformed searches, assume children are expanded left to right. In case of ties, expand in alphabetical order.



Which *one* node will be expanded *next* by each of the following search methods?

A. Depth-First search
B. Greedy Best-First search
C. Uniform-Cost search
D. A* search
E. None of the above

2. Which of the following search algorithms finds an optimal solution?
   A. Breadth first
   B. Hill climbing
   C. Depth first
   D. Greedy search
   E. None of the above

3. Which of these techniques uses randomness to avoid getting trapped in local maxima?
   A. Best first search
   B. Gradient descent
   C. Local beam search
   D. Simulated annealing
   E. None of the above

4. Which of the following is a problem that occurs in hill climbing search?
   A. Cliffs
   B. "Slippery slopes"
   C. Ridges

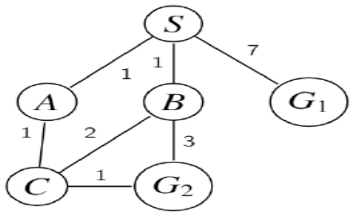
D. Valleys
E. Cycles in the graph

5. A greedy search uses a heuristic function to expand the node that
   A. Appears to be closest to the goal
   B. Is closest to a goal
   C. Is closest to the start state
   D. Is the leftmost node in the search tree
   E. None of the above

6. Which of the following statements is/are true about a heuristic function h?
   (i) If h(n) = h*(n) for all n, then algorithm A* will only expand nodes on the optimal path (ignoring ties).
   (ii) If h is admissible, the smaller h(n) is, the fewer nodes that A* will expand.
   (iii) If h(n) is always less than or equal to the cost of the cheapest path from n to the goal, then A* is guaranteed to find an optimal solution.
   A. Only (i) is true
   B. Only (ii) is true
   C. Only (iii) is true
   D. Both (i) and (iii) are true
   E. All (i), (ii), and (iii) are true

7. Which goal is reached and what is the total cost of the solution found for the following state-space graph when using Breadth-First Search and Uniform-Cost Search (S is the start state, G1 and G2 are the goal states, arcs are bidirectional, no repeated state checking, break any ties alphabetically)?

   A. BFS: G1 (Cost: 7), UCS: G2 (Cost: 4)
   B. BFS: G2 (Cost: 4), UCS: G1 (Cost: 7)
   C. BFS: G2 (Cost: 4), UCS: G2 (Cost: 4)
   D. BFS: G1 (Cost: 7), UCS: G2 (Cost: 3)
   E. BFS: G1 (Cost: 7), UCS: G1 (Cost: 7)



For the next three questions, say we define an evaluation function for a heuristic search problem as: f(n) = (w ∗ g(n)) + ((1 – w) ∗ h(n)) where g(n) is the cost of the best path found from the start state to state n, h(n) is an admissible heuristic function that estimates the cost of a path from n to a goal state, and $0.0 \leq w \leq 1.0$. What search algorithm do you get when:

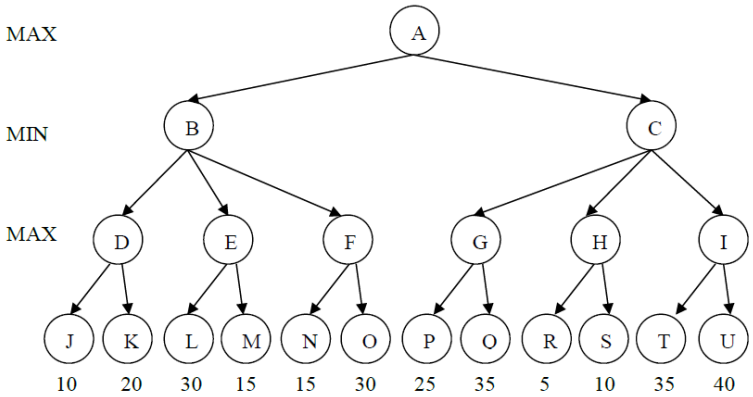
8. w = 0.0
   A. Breadth-First search
   B. Uniform-Cost search
   C. Greedy Best-First search
   D. Algorithm A* search
   E. None of the above

9. w = 0.5
   A. Breadth-First search
   B. Uniform-Cost search
   C. Greedy Best-First search
   D. Algorithm A* search
   E. None of the above

10. w = 1.0
   A. Breadth-First search
   B. Uniform-Cost search
   C. Greedy Best-First search
   D. Algorithm A* search
   E. None of the above Local Search

11. What kind of search best describes what Simulated Annealing does (approximately) if the temperature is very large (i.e., close to ∞) at every iteration?
    A. It will halt immediately and do no search
    B. Breadth-First search
    C. Depth-First search
    D. Hill-Climbing
    E. It will move to a randomly selected successor state at each iteration

12. Consider the following game tree. The root is a maximizing node, and children are visited left to right.



Ignoring the values given above at the leaf nodes, could there possibly exist some set of values at the leaf nodes in the above tree so that Alpha-Beta pruning would do the following. Circle either Yes or No.

(i) Pruning occurs at the arc from D to K. Yes No

(ii) Pruning occurs at the arc from E to M. Yes No

(iii) (iii) [2] Pruning occurs at the arc from C to G. Yes No

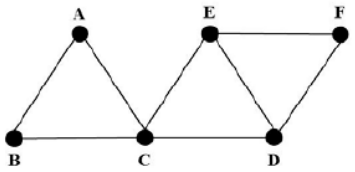
(iv) In the tree above assume that the root node is a MAX node, nodes B and C are MIN nodes, and the nodes D, …, I are not MAX nodes but instead are positions where a fair coin is flipped (so going to each child has probability 0.5). What is the Expectimax value at node A?

13. Which of the following methods is the main way to avoid the horizon effect?
    A. Run Alpha-Beta search with an increasing depth-limit (iterative-deepening search)
    B. When the evaluation function value is frequently changing, look deeper than the depth limit
    C. For each game state, consider only the n best moves (according to the evaluation function) rather than considering all possible moves
    D. Use Expectiminimax to calculate the value of non-terminal game states
    E. None of the above

14. The 8-Queens problem can be represented as an 8-digit vector where the ith digit indicates the row number of the queen in the ith column. Given a state defined by such a vector, a successor function could be defined by randomly selecting a new row for a single queen. Which one of the following actions in a Genetic Algorithm is most similar to the operation defined by this successor function?
    A. Selection
    B. Fitness
    C. Crossover
    D. Mutation
    E. None of the above

15. How would Simulated Annealing perform if the temperature, T, is always very close to 0? Select the most similar one of the following:
    A. A random walk
    B. Hill-Climbing
    C. It would halt immediately
    D. It would never halt
    E. None of the above

Consider the graph representing 6 countries on a map that needs to be coloured using three different colours, 1, 2 and 3, such that no adjacent countries have the same colour. Adjacencies are represented by edges in the graph. We can represent this problem as a Constraint Satisfaction Problem where the variables A to F are the countries, and the values are the colours.



16. The table below shows the current domains of all countries after country B has been assigned value 1 and country F has been assigned value 2. What are the reduced domains of the two countries C and D after applying the Forward Checking algorithm to this state? (The domains of countries A and E may change too but we're not interested in them.)

A	B	C	D	E	F
{1, 2, 3}	{1}	{1, 2, 3}	{1, 2, 3}	{1, 2, 3}	{2}

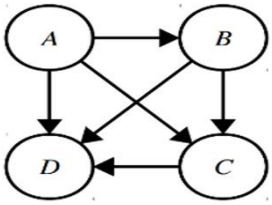
A. C = {1, 2, 3} and D = {1, 2, 3}
B. C = {2, 3} and D = {1, 2, 3}
C. C = {2, 3} and D = {1, 3}
D. C = {2} and D = {1, 3}
E. C = {2} and D = {3}

17. The table below shows the current domains of all countries after country A has been assigned value 2 and country D has been assigned value 3. What are the reduced domains of the two countries C and F after applying the Arc Consistency algorithm (AC-3) to this state? (The domains of countries B and E may change too but we're not interested in them.)

A	B	C	D	E	F
{2}	{1, 2, 3}	{1, 2, 3}	{3}	{1, 2, 3}	{1, 2, 3}

A. C = {1} and F = {1}
B. C = {1} and F = {2}
C. C = {1, 3} and F = {1, 2)
D. C = {1, 3} and F = {1}
E. C = {1, 2} and F = {1, 2}

The following diagram represents a set of people named A, B, C, and D. If there's an arrow from a person x to a person y, then person x likes person y. We'll denote this by writing Likes(x, y). Below is a list of formulas in first-order logic about the picture. In those formulas, the letter P represents the set of all the people. For each formula, determine whether that formula is true or false.



18. $\forall x \in P.\ \forall y \in P.\ (\text{Likes}(x, y) \vee \text{Likes}(y, x))$ True False
19. $\exists x \in P.\ \forall y \in P.\ (\text{Likes}(x, y))$ True False
20. $\exists x \in P.\ \forall y \in P.\ (\neg\text{Likes}(x, y))$ True False

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ANSWER SHEET for Exercise 1

Put a cross in the cell corresponding to your answer. If Yes/No or True/False question, column A corresponds to an answer Yes or True and column B corresponds to No or False as per the question. For Question 12(iv) write the answer in cell A.
The total below is out of 20 but will be scaled down to a mark out of 6.
THIS IS THE SHEET THAT WILL BE MARKED FOR EXERCISE 1. MAKE SURE YOU REPORT YOUR ANSWERS BELOW.

Question N°	A	B	C	D	E	Max Mark
1	E	C	D	G		1.5
2	X					0.5
3			X			0.5
4			X			1
5	X					1
6				X		1
7				X		1
8			X			1
9				X		1
10		X				1
11					X	1
12 (i)		X				1
12 (ii)	X					1
12 (iii)		X				1
12 (iv)	15					1
13		X				1
14				X		1
15		X				1
16			X			1
17	X					1
18		X				0.5
19		X				0.5
20	X					0.5

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Exercise 2: 7 points

Exercise 2: CSP "Four Friends at the Haunted Building"

Four friends—Mustapha, Ali, Rym, and Omar—are moving into the old El Ghoul building, which has exactly three floors:

- Floor 1 = Ground
- Floor 2 = Middle
- Floor 3 = Top

Normally, they'd pick any floor they like—even share rooms—but during last night's ghost tour, they heard the following warnings:

1. **Mustapha and Ali awaken the Ghost of El Ghoul building whenever they're together.** They must live on *different floors*.
2. **Rym only gets scared by Ali if they're together on floor 1 or 3.** So, if Rym and Ali end up on the same floor, it must be floor 2.
3. **If Rym and Ali end up on different floors, then at least one of them must live on the top floor.**
4. **Omar is convinced the building is haunted.** He refuses to have any roommates and must have a floor *all to himself*.
5. **Omar insists on having the best view over the cemetery.** He must live *strictly above* Rym.

When morning comes, they discover that the keys are missing from the lobby. No one can move around to check who's living where.

Tasks:

1. Formulate this problem as a Constraint Satisfaction Problem (CSP) so that a computer can determine all valid floor assignments. Clearly define the following components:
   a) **Variables**
   b) **Domains**
   c) **Constraints:** Use simple mathematical or logical notation to express the constraints. Enumerate each constraint (e.g., C1, C2, etc.).

2. You decide to give Ali a room on the first floor. Using **Forward Checking**, determine the remaining possible floors for **Rym**. Justify your answer based on the constraints ?

3. **Using only the constraints that involve Omar**, apply the AC-3 algorithm to enforce arc consistency. Show the steps of the algorithm in the table below. After completing the algorithm (in the below table), report the possible floors that Omar can occupy.

Steps:

Step 0 - Initial Arcs:

queue ={.....}

Step 1 - Loop through arcs:

Arc to be revised	Concerned Constraint number	Domains before revision	Domains after revision	Queue after revision

Final Possible Floors for Omar: ……..…..

Solution:

Q1 Formulation as CSP (2.25 pts):

- **Variables:** Mustapha (M), Ali(A), Rym(R), and Omar(O)
- **Domains:** {1, 2, 3} (floors)
- **Constraints:**

1. **C1:** $M \neq A$
2. **C2:** If $R = A$ then $R = A = 2$
3. **C3:** If $R \neq A$ then $R = 3$ or $A = 3$
4. **C4:** O is alone: $O \neq M$, $O \neq A$, $O \neq R$
5. **C5:** $O > R$

Q2 Forward Checking R =? if A = 1 (1 pts):

- R's domain must satisfy C2 & C3 given A = 1.
- C2 says if R=1 then both must be in 2nd floor (**impossible**) ⇒ remove 1 from R possible values (D(R) = {2, 3})
- C3 says if R≠1 then R = 3 or A = 3, ⇒ since A can't be 3, then R must be 3
  Answer: {3}

Q3 AC-3 on O steps (3.75 pts):

- Varaibles: **M, A, R, O** for (Mustapha, Ali, Rym, and Omar)
- Initial Domains: D(M) = D(A) = D(R) = D(O) = {1,2,3}
- Binary Constraints (for Omar (O)):

  - **C1**: O ≠ M
  - **C2**: O ≠ A
  - **C3**: O ≠ R
  - **C4**: O > R

Step 0: Initialize queue with all arcs

The queue starts with all arcs, including both directions for each constraint:

Q = {(O,A),(A,O),(O,M),(M,O),(O,R),(R,O)}.

Step 1: Loop through arcs:

Step	Arc Revised	Constraint(s)	Domains before revision	Domains after revision	Queue after revision
0	—	—	D(M)=D(A)=D(R)=D(O)={1,2,3}	—	(O,A), (A,O), (O,M), (M,O), (O,R), (R,O)
1	**(O,A)**	C4: O≠A	D(O)={1,2,3}, D(A)={1,2,3}	no change	(A,O),(O,M),(M,O),(O,R),(R,O)
2	**(A,O)**	C2: O≠A	D(A)={1,2,3}, D(O)={1,2,3}	no change	(O,M),(M,O),(O,R),(R,O)
3	**(O,M)**	C1: O≠M	D(O)={1,2,3}, D(M)={1,2,3}	no change	(M,O),(O,R),(R,O)
4	**(M,O)**	C1: O≠M	D(M)={1,2,3}, D(O)={1,2,3}	no change	(O,R),(R,O)
5	**(O,R)**	C3: O≠R C4: O>R	D(O)={1,2,3}, D(R)={1,2,3}	**D(O)={2,3}**	(R,O),**(A,O),(M,O)**
6	**(R,O)**	C3: O≠R C4: O>R	D(R)={1,2,3}, D(O)={2,3}	**D(R)={1,2}**	(A,O),(M,O)
7	**(A,O)**	C2: O≠A	D(A)={1,2,3}, D(O)={2,3}	no change	(M,O)
8	**(M,O)**	C1: O≠M	D(M)={1,2,3}, D(O)={2,3}	no change	*Ø (no change, algorithm terminates)*

Final Possible Floors for Omar: {2,3}

---------- **AC3** ----------

• **Initialize**
Add every directed arc (X→Y) for each constraint in the CSP.
(If you split each constraint into its own arc, each gets its own entry; if you combine multiple constraints on the same pair, you just have one combined arc.)

• **Loop until no arcs left**
• Remove an arc (X→Y) from the queue.
• Call **Revise** on it.

• **Revise(X, Y)**
• For each value x in Dom(X):
– If there's **no** y in Dom(Y) satisfying **all** constraints between X and Y, remove x from Dom(X).
• If Dom(X) shrank, **enqueue** every arc ending in X and its starting variable is different that Y.

• **Finish**
When the queue is empty, every value in each domain has support in all neighbours-CSP is now arc-consistent.

Exercise 3: 8 points

A. Rewrite the following statements in first-order logic. We will deal with a simple setting where we only have three objects in the world: Sanaa, Yacine, and Badri. Choose appropriate predicates and constants.

1. Sanaa is an employee.
2. Yacine is a boss.
3. Sanaa works.
4. Yacine works.
5. Sanaa gets paid.
6. Yacine gets paid.
7. Badri is not an employee.
8. Badri doesn't work.
9. Badri doesn't get paid
10. All employees work.
11. All employees are paid.
12. Bosses are not employees.
13. If someone doesn't work, they are not paid.
14. There exists at least one boss who is paid.
15. There exists someone who is neither an employee nor a boss.
16. Everyone who is paid is either a boss or an employee.
17. There exists an employee who works and gets paid.

A. Transform the above FOL sentences into CNF, showing all the steps.

B. Apply the Resolution inference rule to prove the conclusion (statement 17) from the remaining statements 1 to 16.

C. We know from the lectures that one technique we can use to perform inference with our knowledge base is to propositionalise the statements of first-order logic into statements of propositional logic. Practice this by propositionalising statement (10) from the above knowledge base.

D. If we translated the statement "Anyone who is not a boss either works or does not get paid" into first-order logic and added it to our knowledge base, how would the size of the set of valid models representing our knowledge base change, and why? Give a formal proof.

E. Using only our original knowledge base, we want to answer the question "Does every employee work?" We first translate the sentence "every employee works" into first-order logic as statement 10. Determine the answer to our query by considering the following questions of satisfiability:

i. Fill in the following table with T for true and F for false.

x	Employee(x)	Boss(x)	Works(x)	Paid(x)
Alice				
Bob				
Carol				

ii. From the previous table, is KB ∪ ¬10 satisfiable? Explain.

iii. Fill in the following table with T for true and F for false.

x	Employee(x)	Boss(x)	Works(x)	Paid(x)
Alice				
Bob				
Carol				

iv. Is KB ∪ 10 satisfiable? Explain

v. Based on your answers to the previous two parts, does our knowledge base entail 10, contradict 10, or neither? Explain.

A. Rewrite the following statements in first-order logic. We will deal with a simple setting where we only have three objects in the world: Sanaa, Yacine, and Badri. Choose appropriate predicates and constants. 2.75 pts

1. Sanaa is an employee. 2. Yacine is boss.	Employee(Sanaa) Boss(Yacine)	0.25 pts in total
3. Sanaa works. 4. Yacine works.	Works(Sanaa) Works(Yacine)	0.25 pts in total
5. Sanaa gets paid. 6. Yacine gets paid.	Paid(Sanaa) Paid(Yacine)	0.25 pts in total
7. Badri is not an employee. 8. Badri doesn't work. 9. Badri doesn't get paid	¬Employee(Badri) ¬Works(Badri) · ¬Paid(Badri)	0.25 pts in total

10. All employees work. 11. All employees are paid.	$\forall x \ (Employee(x) \rightarrow Works(x))$ $\forall x \ (Employee(x) \rightarrow Paid(x))$	0.25 pts each
12. Bosses are not employees.	$\forall x \ (Boss(x) \rightarrow \neg Employee(x))$	0.25 pts
13. If someone doesn't work, he/she is not paid.	$\forall x \ (\neg Works(x) \rightarrow \neg Paid(x))$	0.25 pts
14. There exists at least one boss who is paid.	$\exists x \ (Boss(x) \wedge Paid(x))$	0.25 pts
15. There exists someone who is neither an employee nor a boss.	$\exists x \ (\neg Employee(x) \wedge \neg Boss(x))$	0.25 pts
16. Everyone who is paid is either a boss or an employee (i.e., paid people belong to one of those categories).	$\forall x \ Paid(x) \rightarrow (Boss(x) \wedge \neg Employee(x)) \vee (\neg Boss(x) \wedge Employee(x))$	0.25 pts
17. There exists an employee who works and gets paid.	$\exists x \ (Employee(x) \wedge Works(x) \wedge Paid(x))$	0.25 pts

Transform these sentences into CNF, showing all the steps. 2 pts

Solution:

$\forall x \ (Employee(x) \rightarrow Works(x))$ $\forall x \ (Employee(x) \rightarrow Paid(x))$	$\neg Employee(x) \vee Works(x)$ $\neg Employee(x) \vee Paid(x)$	0.25 pts if both are correct. If one is incorrect then 0 to both.
$\forall x \ (Boss(x) \rightarrow \neg Employee(x))$	$\neg Boss(x) \vee \neg Employee(x)$	0.25 pts if the two conversions are

∀x (¬Works(x) → ¬Paid(x))	Works(x) ∨ ¬Paid(x)	correct; if either one is incorrect, then 0.
∃x (Boss(x) ∧ Paid(x))	Skolemization: Introduce constant A: `Boss(A) ∧ Paid(A)` a. CNF: `Boss(A)` b. CNF: `Paid(A)`	0.25 pts
∃x (¬Employee(x) ∧ ¬Boss(x))	Skolemization Introduce constant C: `¬Employee(C) ∧ ¬Boss(C)` a. CNF: `¬Employee(C)` b. CNF: `¬Boss(C)`	0.25 pts
∀x Paid(x) → (Boss(x) ∧ ¬Employee(x)) ∨ (¬Boss(x) ∧ Employee(x))	Eliminate implication: ∀x ¬ Paid(x) ∨ [(Boss(x) ∧ ¬Employee(x)) ∨ (¬Boss(x) ∧ Employee(x))] Distribute ∨ over ∧ inside the [ ]: ∀x ¬Paid(x) ∨ **[(Boss(x) ∨ ¬Boss(x))** ∧ (Boss(x) ∨ Employee(x)) ∧ (¬Employee(x) ∨ ¬Boss(x)) ∧ **(¬Employee(x) ∨Employee(x)]** ∀x ¬Paid(x) ∨ **[**(Boss(x) ∨ Employee(x)) ∧ (¬Employee(x) ∨ ¬Boss(x))] Distribute external ∨ over internal ∧: ∀x (¬Paid(x) ∨ Boss(x) ∨ Employee(x)) ∧ (¬Paid(x) ∨ ¬Employee(x) ∨ ¬Boss(x))] Remove ∀x and eliminate ∧ yielding 2 clauses in CNF: • ¬Paid(x)∨(Boss(x)∨Employee(x) (16.a) • ¬Paid(x)∨¬Employee(x)∨¬Boss(x) (16.b)	0.5 pts if all steps and result are correct; otherwise give either 0.25 or 0 depending on how much is done.
∃x (Employee(x) ∧ Works(x) ∧ Paid(x))	Skolemization Introduce constant B: `Employee(B) ∧ Works(B) ∧ Paid(B)` a. CNF: `Employee(B)` b. CNF: `Works(B)` c. CNF: `Paid(B)`	0.5 pts if all correct; otherwise, 0.

A. Apply the Resolution inference rule to prove the conclusion (statement 17) from the remaining statements 1 pt (If the explanation that we need to negate the goal and reach a contradiction

to prove it is not given, deduct 0.25 pts from the total obtained as per the following steps given in the table)

Step 1: Negate the original statement 17 (i.e. what we want to prove from the KB): ∃x (Employee(x) ∧ Works(x) ∧ Paid(x))	Negated: ¬∃x (Employee(x) ∧ Works(x) ∧ Paid(x)) ≡ ∀x (¬(Employee(x) ∧ Works(x) ∧ Paid(x))) ≡ ∀x (¬Employee(x) ∨ ¬Works(x) ∨ ¬Paid(x)) We drop the universal quantifier and get the clause: ¬Employee(x) ∨ ¬Works(x) ∨ ¬Paid(x)	0.5

We'll try to derive a contradiction by adding this negated goal ¬Employee(x) ∨ ¬Works(x) ∨ ¬Paid(x) to the KB:

Knowledge Base (Without Statement 17)

We'll use relevant KB clauses and number the clause as indicated below:

¬Employee(x) ∨ Works(x) // clause 10
¬Employee(x) ∨ Paid(x) // clause 11
Works(B) // clause 12
Employee(B) // clause 13
Paid(B) // clause 14

Negated goal: ¬Employee(x) ∨ ¬Works(x) ∨ ¬Paid(x) // ¬Goal

Unify clause 1 with ¬Employee(x) in clause ¬Goal with substitution {x/Sanaa}	Produced clause 18: ¬Works(Sanaa) ∨ ¬Paid(Sanaa)	0.5 pts
Unify clause 3 with clause 18	Produced clause 19: ¬Paid(Sanaa)	
Unify clause 5 with ¬Works(b) in clause 19	Produced empty clause so contradiction	

Therefore ∃x (Employee(x) ∧ Works(x) ∧ Paid(x)) is derived from the KB.

B. We know from class that one technique we can use to perform inference with our knowledge base is to propositionalize the statements of first-order logic into statements of propositional logic. Practice this by propositionalizing statement (10) from our knowledge base. 0.5 pts (if the 3 sentences given; 0.25 pts if 2 of the 3 sentences are given; 0 if only one sentence is given)

Solution

We restrict the domain to the constants:

Domain D = {Sanaa, Yacine, Badri}

To propositionalize the universally quantified statement ∀x (Employee(x) → Works(x)), we **instantiate** it for each constant:

Employee(Sanaa) → Works(Sanaa)
Employee(Yacine) → Works(Yacine)
Employee(Badri) → Works(Badri)

C. If we translated the statement "Anyone who is not a boss either works or does not get paid" into first-order logic and added it to our knowledge base, how would the size of the set of valid models representing our knowledge base change, and why?

0.5 pts

Solution

The size of the set of valid models would either decrease or stay the same, because the new statement introduces an additional constraint that some previously valid models might not satisfy.

Why?

Because any model that previously allowed a person to be not a boss, not work, **and** get paid would now **violate** the new sentence and be **excluded** from the set of valid models.

D. Using only our original knowledge base, we want to answer the question "Does everyone work?" We first translate the sentence "everyone works" into first-order logic as statement 10. Determine the answer to our query by considering the following questions of satisfiability:

i. Fill in the following table with T for true and F for false. 0.25 pts

x	Employee(x)	Boss(x)	Works(x)	Paid(x)
Sanaa	T	F	T	T
Badri	F	F	F	F
Yacine	F	T	T	T

ii. From the previous table, is KB ∪ ¬10 satisfiable?

Solution: No 0.25 pts

From the previous table we see that **KB ∪ ¬10 is not satisfiable** because ¬10 says that "Not every employee works", yet Sanaa, the only employee, does work (so every employee works) and the KB is still true.

iii. Fill in the following table with T for true and F for false. 0.25 pts

iv. Is KB ∪ 10 satisfiable?

x	Employee(x)	Boss(x)	Works(x)	Paid(x)

Sanaa	T	F	T	T
Badri	F	F	F	F
Yacine	F	T	T	T

Solution: Yes 0.25 pts

From the previous table we see that **KB ∪ 10 is satisfiable** because there is a model where **every employee works** (in this case Sanaa only), and the KB is still true. So **10 is not contradicted** by KB.

v. Based on your answers to the two parts, does our knowledge base entail 10, contradict 10, or neither? 0.25 pts

Solution:

Part A: We showed that **KB ∪ ¬10 is not satisfiable** → There is a model where "**not every employee works"** is false (**every employee does work**), and the KB is still true. So **¬10 is contradicted** by KB. So 10 is entailed by KB.

Part B: We also showed that **KB ∪ 10 is satisfiable** → There is a model where "**every employee works"** is true (Sanaa being the only employee and she works), and the KB is still true. So **10 is not contradicted** by KB.

This means **statement 10 is logically entailed** from the KB.